# Do Now · retrieval
## Karnaugh maps and simplification

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Simplify ¬(A ∨ B).

### Q2

Keys 7 and 12 hash to index 2 in a table. What is that called?

### Q3

What is 1 XOR 1?

### Q4

If A changes from 0 to 1 while B is always 1, which variable is constant?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Read the labels — Rows represent A, and columns use Gray order BC: 00, 01, 11, 10.

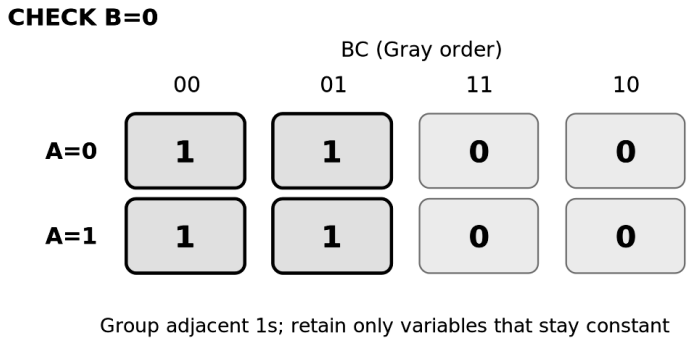
## Core idea 2

Group neighbouring ones — The four 1s form one 2×2 rectangle. Adjacency across rows is allowed.

## Core idea 3

Keep what is constant — A and C change through the group, but B remains 0. So the simplified result is ¬B.

**Key visual · complete concept**




Group adjacent 1s; retain only variables that stay constant

## Annotate the key visual

Explain which labels or features show the main process.

# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

Karnaugh map — Grid for simplifying Boolean functions using adjacent combinations.

Gray code — Ordering in which adjacent binary labels differ by one bit.

group — A rectangle containing 1, 2, 4, 8, … adjacent 1-cells.

constant variable — A variable with the same value in all cells of a group.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

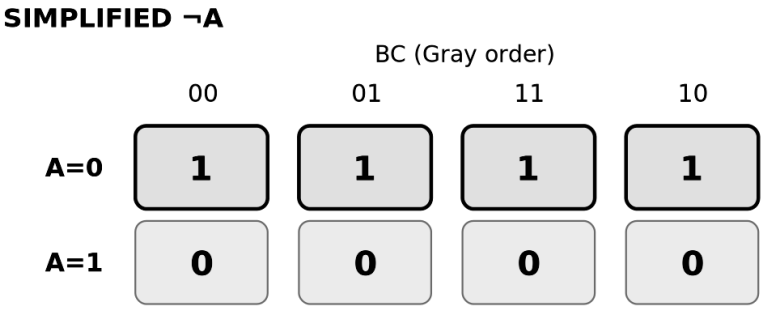
## Problem

Four 1s fill the entire A=0 row. Simplify the output.

## Follow the worked method

1. Predict and explain — B and C each vary across the row; neither is needed in the expression.
2. Check and extend — A stays 0 in the whole group. The simplified term is ¬A.

**Source model · final illustrated step**




Group adjacent 1s; retain only variables that stay constant

## Why this method works

Identify the decision, rule or invariant that makes the method correct.

# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict and explain
A changes between rows, and C changes between columns 11 and 10.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Why do we group neighbouring 1s in a Karnaugh map?

## Question 2

The whole A=1 row contains four 1s. Which simplified term remains?

Check that you have used the technical terms and explained your decisions.

# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Columns BC=00 and BC=10 are joined by wraparound adjacency. Four 1s occupy both these columns. Simplify.

Check that you have used the technical terms and explained your decisions.

# Exit Ticket · reflect and retrieve

## Exit Ticket

A group of four 1s can remove two changing variables.

## Use key vocabulary from memory

1. Explain Karnaugh map in your own words.

2. Explain Gray code in your own words.

3. Explain group in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.